

Verdier Duality

Let X, Y be loc. compact top. spaces, with categories of sheaves of abelian groups $\text{Sh}X, \text{Sh}Y$. Let $f: X \rightarrow Y$. For $A \in \text{Sh}X$, define its direct image with proper support $f_! A \in \text{Sh}Y$ by $f_! A(U) = \{s \in \Gamma(f^{-1}(U)) : f|_{\text{supp } s} \text{ is proper}\}$.

Its stalks are $(f_! A)_y \cong \Gamma_c(f^{-1}(y), A|_{f^{-1}(y)})$.

We say that $A \in \text{Sh}X$ is c-soft if $\Gamma(X, A) \rightarrow \Gamma(K, A)$ for any compact $K \subset X$. Note that injective sheaves are c-soft. Also, $f_! A$ is acyclic for any c-soft sheaf A .

Let $R^i f_! A = H^i(Rf_! A)$. Then

$$\begin{aligned} (R^i f_! A)_y &= H^i(Rf_! A)_y \cong H^i(\Gamma_c(f^{-1}(y), A|_{f^{-1}(y)})) \\ &= H^i(R\Gamma_c(f^{-1}(y), A)) =: H_c^i(f^{-1}(y), A). \end{aligned}$$

Note that if $i: X \hookrightarrow Y$ is a closed or open inclusion then, calculating the stalks $i_!$ is exact. If i is closed then $i_! = i_*$.

Inverse image with proper support

We want to construct a right adjoint to $f_!$. It will not exist on the level of sheaves.

Let us analyse why:

Let $i: U \hookrightarrow X$ be an open inclusion. We have for any $V \subset X$ open, $F \in \text{Sh}X$,

$$\begin{aligned} \Gamma(U, i^! F) &= \text{Hom}_U(Z_U, i^! F) = \Gamma(U, i_* \text{Hom}(Z_U, i^! F)) \\ &= \Gamma(U, \text{Hom}(i_! Z_U, F)) \\ &= \text{Hom}(i_!^* Z_U, i^* F) = \Gamma(U, i^* F). \end{aligned}$$

Hence, $i^! = i^*$. Now for $f: X \rightarrow Y$,

$$\begin{aligned} \Gamma(U, f^! g) &= \Gamma(U, i^* f^! g) = \Gamma(U, i^! f^! g) \\ &= \Gamma(U, (f|_U)^! g) = \text{Hom}(Z_U, (f|_U)^! g) \\ &= \Gamma(U, \text{Hom}(Z_U, (f|_U)^! g)) \\ &= \Gamma(Y, (H|_U)_* \text{Hom}(Z_U, (f|_U)^! g)) \\ &= \Gamma(Y, \text{Hom}(H|_U)_! Z_U, g) \\ &= \text{Hom}((f|_U)_! Z_U, g) = \text{Hom}(f_!(i_! Z_U), g). \end{aligned}$$

But this is not a sheaf in general.

To remedy this, let $K \in \text{Sh}X$ be a c-soft flat sheaf. We get

$$\text{Hom}(f_!(F \otimes K), g) = f_* \text{Hom}(F, f_K^! g),$$

where $f_K^! g: U \rightarrow \text{Hom}(f_!(i_! K), g)$.

Now fix a c-soft flat resolution K^* of Z_X . We get

$$\text{Hom}^*(f_!(F \otimes K^*), g) = f_* \text{Hom}^*(F, f_K^! g).$$

Then $f_K^!$ induces a functor $f^!$ in $D^b X$ by setting

$$f^! F = f_K^! I^*$$

for an injective resolution I^* of F .

Thm (Verdier Duality): We have

$$R\text{Hom}^*(Rf_! F, g) = Rf_* R\text{Hom}^*(F, f^! g).$$

Note that

$\text{Hom}(F, f_* g) = f_* \text{Hom}(f^* F, g)$
ie (f^*, f_*) is an adjoint pair,
while here, $(f_!, f^!)$ is.

Proof: We have $f^! g = f_K^! I^*$

for an injective resolution I^* . Check that $f_K^! I^*$ is injective, so that

$$R\text{Hom}^*(F, f_K^! I^*) = \text{Hom}^*(F, f_K^! I^*).$$

Check that this is flasque, so that $Rf_* \text{Hom}^*(F, f_K^! I^*) = f_* \text{Hom}^*(F, f_K^! I^*)$.

For the LHS, $F \otimes K^*$ is a c-soft resolution of F , so that

$$Rf_! F = f_!(F \otimes K^*).$$

Hence, $R\text{Hom}^*(Rf_! F, g) = \text{Hom}^*(f_!(F \otimes K^*), g)$,

and the statement follows from the above.

Cor: $\text{Hom}_{D^b(Y)}(Rf_! F, g) = \text{Hom}_{D^b(X)}(F, f^! g)$.

Proof: For any $A, B \in \mathcal{I}^*$, use that

$$\begin{aligned} \text{Hom}_D(A, B) &= \text{Hom}_D(A, \mathcal{I}^*) = \text{Hom}_X^*(A, \mathcal{I}^*) \\ &= H^0 \Gamma(X, \text{Hom}^*(A, \mathcal{I}^*)) = H^0 \Gamma(X, R\text{Hom}^*(A, B)). \end{aligned}$$

The dualising functor

Let $D_X = \pi^! \mathbb{Z}$, where $\pi: X \rightarrow \text{pt}$. We have

$$\begin{aligned} \Gamma(U, D_X) &= \Gamma(U, \pi^! \mathbb{Z}) = \Gamma(U, f_K^! \mathbb{Z}) = \text{Hom}^*(f_!(i_! K|_U), \mathbb{Z}) \\ &= \text{Hom}(\Gamma_c^*(X, i_! K|_U), \mathbb{Z}). \end{aligned}$$

Check that $\pi^!$ takes injectives to injectives, so that D_X is injective.

Def: The functor $D_X = R\text{Hom}^*(\cdot, D_X)$ is the dualising functor.

Let $i: U \hookrightarrow X$. Then

$$\begin{aligned} \Gamma(U, D_X F) &= \Gamma(U, \text{Hom}^*(F, D_X)) \\ &= \Gamma(U, \text{Hom}^*(F, f_K^! \mathbb{Z})) \\ &= \text{Hom}^*(i^* F, i^* f_K^! \mathbb{Z}) \\ &= \text{Hom}^*(i^* F, (\pi|_U)^! \mathbb{Z}) \quad i^* = i^! \\ &= \text{Hom}^*((\pi|_U)_! (F \otimes K|_U), \mathbb{Z}) \\ &= \text{Hom}^*(\Gamma_c^*(U, F \otimes K), \mathbb{Z}). \end{aligned}$$

Prop: Let $f: X \rightarrow Y$, $F \in D^b(X)$, $G \in D^b(Y)$. Then

$$D_Y(Rf_! F) \cong Rf_* D_X F, \quad D_X(f^* G) \cong f^! D_Y G.$$

Proof: First note that $D_X = \pi_X^! \mathbb{Z} = (f_* \pi_Y)^! \mathbb{Z} = f^! D_Y$.

Hence,

$$\begin{aligned} D_Y(Rf_! F) &= R\text{Hom}(Rf_! F, D_Y) \\ &= Rf_* \text{Hom}(F, f^! D_Y) \\ &= Rf_* \text{Hom}(F, D_X) = Rf_* D_X F. \end{aligned}$$

The other formula follows from $f^! R\text{Hom}(G, H) = R\text{Hom}(f^* G, f^! H)$.

Let M be a manifold. Let $\mathcal{O}_M$ be its orientation sheaf $U \mapsto \text{Hom}(H_c^*(U; \mathbb{Z}), \mathbb{Z})$. We claim that we have

$$(1) \quad D_M \cong \mathcal{O}_M[n].$$

For $U = \mathbb{R}^n$,
 $R\Gamma(U, D_X) = \text{Hom}_{D^b}(\mathbb{Z}_U, f^! \mathbb{Z})$
 $= \text{Hom}_{D^b}(\mathbb{R}_f, \mathbb{Z})$
 $= R\Gamma_c^*(U, \mathbb{Z})^*$

Recall that

$$(2) \quad \text{Hom}_{D^b(X)}(Rf_! F, G) = \text{Hom}_{D^b(Y)}(F, f^! G).$$

Taking $F = k_X[i]$, $G = k$, $\pi: M \rightarrow \text{pt}$, the LHS is

$$\begin{aligned} \text{Hom}_{D^b(M)}(R\pi_! k_X[i], k) &= H^0 \text{Hom}_{D^b(\text{pt})}(\pi_! \mathbb{Z}[i], k) \\ &= H_c^i(X; k_X)^*. \end{aligned}$$

The RHS becomes

$$\begin{aligned} \text{Hom}_{D^b(X)}(k_X[i], \pi^! k) &= H^0 \text{Hom}_{\mathcal{O}_M}(k_X[i], \mathcal{O}_M[n]) \\ &= H^{n-i}(\text{Hom}(k_X, k_X)) \quad \text{if } \mathcal{O}_M \cong k_X, \text{ i.e. } M \text{ is orientable} \\ &= H^{n-i}(X; k_X). \end{aligned}$$

Intersection homology

Here, all spaces are paracompact and Hausdorff. We first define stratified spaces inductively: A 0-dimensional stratified pseudomanifold is a countable collection of points.

An n -dimensional stratified space X has a filtration

$$X = X_n \supset X_{n-1} = X_{n-2} \supset \dots \supset X_0 \supset \emptyset$$

by closed subsets, s.t.

- $X_{n-k} \setminus X_{n-k-1}$ is an $(n-k)$ -dimensional manifold,
- $X \setminus X_{n-2}$ is dense,
- $\forall x \in X_{n-k} \setminus X_{n-k-1}$ there is a nbhd U of x and a compact stratified pseudomanifold L of dimension $k-1$ with a stratum-preserving homeomorphism $U \xrightarrow{\cong} \mathbb{R}^{n-k} \times \mathbb{C}^0 L$.

open cone

